

WASP-12b

R-0176

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_250126 · created 2026-07-16T18:36:03+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460701.6909 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.045
Transit depth (Rp/Rs)^2	2.38 +/- 1.37 [%]
Orbital Inclination (inc)	86.74 +/- 2.95
Airmass coefficient 1 (a1)	0.083 +/- 0.068
Airmass coefficient 2 (a2)	2.36 +/- 1.69
Scatter in the residuals of the lightcurve fit is	103.13 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	5.75
Transit Duration (day)	0.1094 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.154 $\pm$ 0.045 vs 0.1178 (deviation 0.8 σ)
Duration fit vs published	0.1094 d vs 0.125 d (deviation 4.73 σ)
Mid-time O-C	0.6 min
Depth SNR	3.4
Residual scatter	103.13 %

Light curve

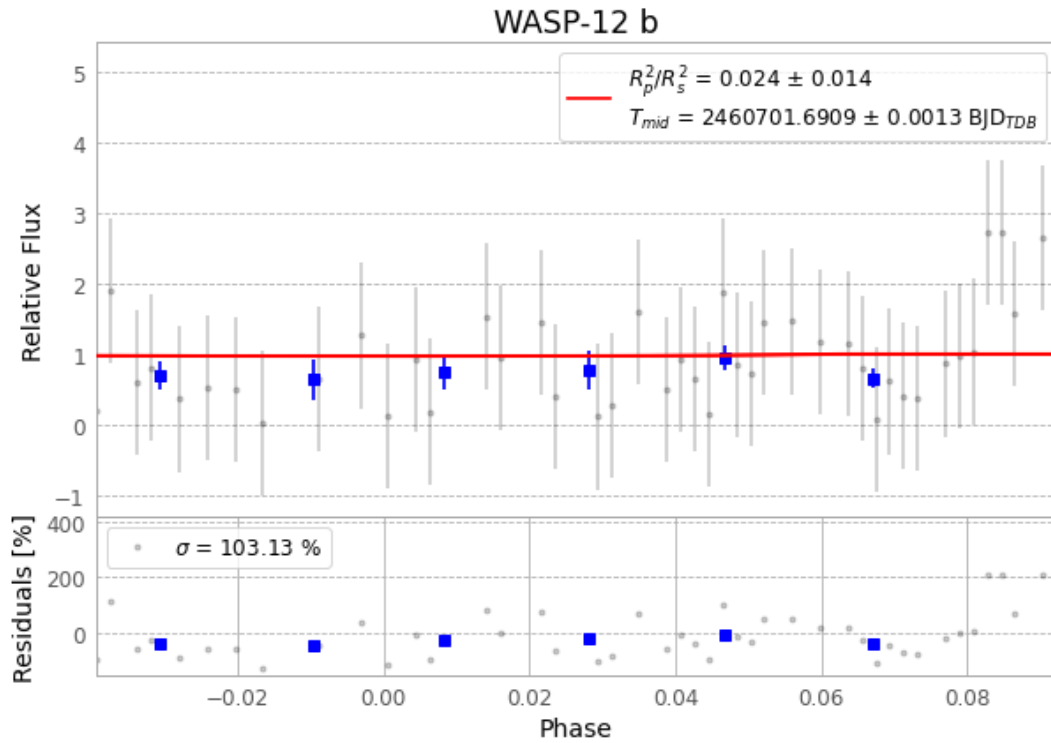


Figure 1 — FinalLightCurve_WASP-12 b_2025-01-25.png (SHA-256 9736d51f88445a37...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [87, 142], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9332a41fd580362d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f67d25a

Data provenance

78 FITS frames (51 MB), WASP-12250126030915.FITS → WASP-12250126070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250126.log (b6aed2d475a3...), FinalLightCurve_WASP-12 b_2025-01-25.png (9736d51f8844...), FinalLightCurve_WASP-12 b_2025-01-25.csv (c6866a764c33...)

Compute resources

Wall time 215.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 18:36Z.